



Analog IC Design Implementation and Verification
Week 5 – Comparators, Stability and Common-Mode Response

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Introduction: Comparators

The comparator compares two voltages and is essentially a 1-bit analog-digital converter:

- **Input:** analog
- **Output:** digital

Types

- Open-loop (cascade of broadband stages)
- Regenerative (use of *positive feedback* latches)
- Combination of open-loop and regenerative comparators

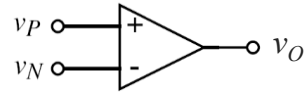


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Comparator Characteristics

Static characteristics

- Gain
- Output high and low states
- Input resolution
- Offset
- Noise, CMRR, PSRR and so forth



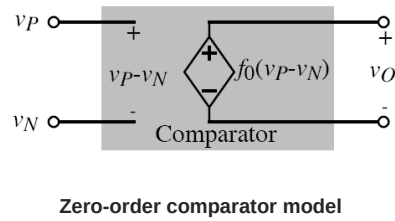
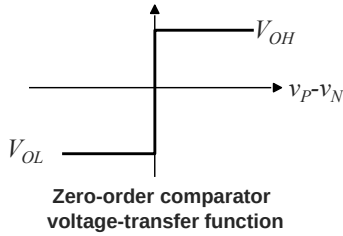
Comparator circuit symbol

Dynamic characteristics

- Propagation delay
- Slew rate

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Static Characteristics: Zero-Order Model for Comparators

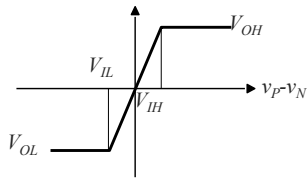


$$\text{Gain} = A_v = \lim_{\Delta V \neq 0} \frac{V_{OH} - V_{OL}}{\Delta V}$$

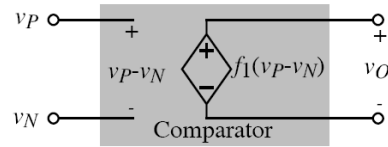
Note: Many behavioral models use the hyperbolic-tangent function to approximate $V_{out}(V_{in})$.

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Static Characteristics: First-Order Models



First-order comparator
voltage-transfer function



First-order
comparator model

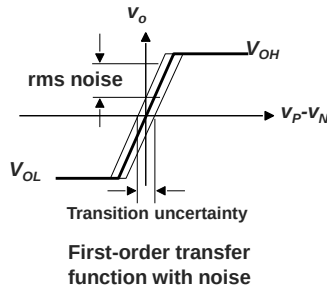
$$f_1(v_P - v_N) = \begin{cases} V_{OH} & \text{for } (v_P - v_N) > V_{IH} \\ A_v(v_P - v_N) & \text{for } V_{IL} < (v_P - v_N) < V_{IH} \\ V_{OL} & \text{for } (v_P - v_N) < V_{IL} \end{cases}$$

$$\text{Gain} = A_v = \frac{V_{OH} - V_{OL}}{V_{IH} - V_{IL}}$$

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Static Characteristics: Comparator Noise

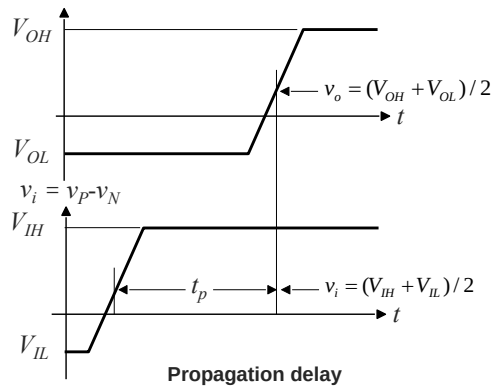
- Noise of a comparator is modeled as if the comparator is biased in the transition region, such as an amplifier.



- Noise leads to an uncertainty in the transition region, which causes jitter or phase noise. The relation between time jitter and voltage uncertainty is defined by signal slew.

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Dynamic Characteristics: Propagation Delay Time



$$\text{Propagation delay time} = \frac{\text{Rising propagation delay time} + \text{falling propagation delay time}}{2}$$

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Multistage Open-Loop Comparators

- Cascaded, actively loaded stages
 - Useful for non-time-critical applications.
 - Response time is a function of history, overdrive, and preoverdrive conditions.
- Cascaded resistively loaded stages
 - Higher speed applications
 - Useful for asynchronous (nonclock) applications



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Dynamic Characteristics: Comparator Slew Rate

If the rate of rise or fall of a comparator becomes input independent, the dynamics can be modeled by a slew rate. Slew rate comes from this basic relationship:

$$i = C \frac{dv}{dt}$$

where i is the current through a capacitor and v is the voltage across it.

If the current becomes limited, then the voltage rate becomes limited. Therefore, for a comparator that is slew-rate limited, you have the following:

$$t_p = \Delta T = \frac{\Delta V}{SR} = \frac{V_{OH} - V_{OL}}{2 \cdot SR} \quad t_p = T = \frac{V_{OH} - V_{OL}}{2 \cdot SR}$$

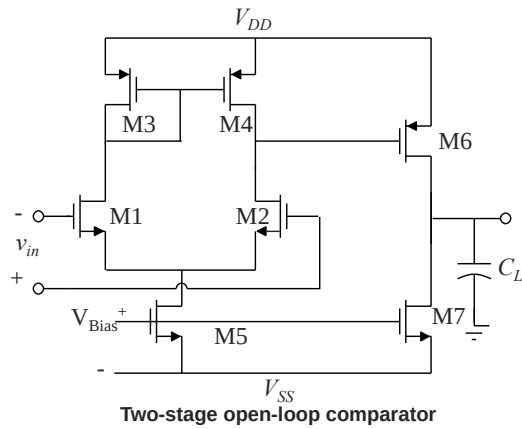
where SR = slew rate of the comparator.



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Two-Stage Open-Loop Active-Load Comparators

Use a high-gain stage to drive outputs between V_{OH} and V_{OL} for very small input voltage changes.



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Active-Load Comparators (continued)

Comparator performance

- Maximum output voltage

$$V_{OH} = V_{DD} - (V_{DD} - V_{G6}(\min) - |V_{TP}|) \left[1 - \sqrt{1 - \frac{8I_7}{\beta_6(V_{DD} - V_{G6}(\min) - |V_{TP}|)^2}} \right]$$

- Minimum output voltage

$$V_{OL} = V_{SS}$$

- Small-signal voltage gain

$$A_v(0) = \left(\frac{g_{m1}}{g_{ds2} + g_{ds4}} \right) \left(\frac{g_{m6}}{g_{ds6} + g_{ds7}} \right)$$

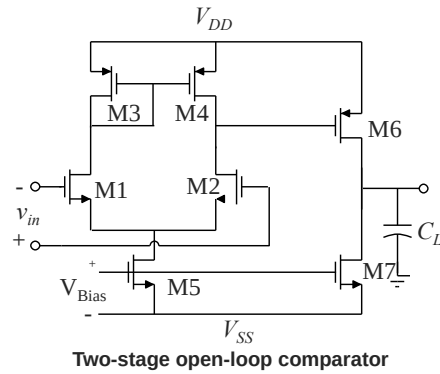
- Poles

- Input $p_1 = \frac{-1}{C_I(g_{ds2} + g_{ds4})}$

- Output $p_2 = \frac{-1}{C_{II}(g_{ds6} + g_{ds7})}$

- Frequency response

$$A_v(s) = \frac{A_v(0)}{\left(\frac{s}{p_1} + 1 \right) \left(\frac{s}{p_2} + 1 \right)}$$

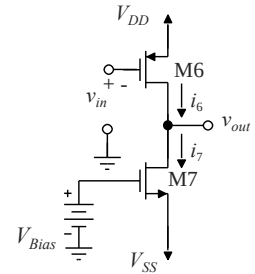


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Output Stage Analysis

- **Inverter trip point:** To determine the propagation delay time, you must know when the second stage of the two-stage comparator begins to turn on.
- Assume that M6 and M7 are saturated. (It is known that the steepest slope occurs for this condition.)
- Equate i_6 to i_7 and solve for v_{in} which becomes the trip point:

$$\therefore v_{in} = V_{TRP} = V_{DD} - |V_{TP}| - \sqrt{\frac{K_N(W_7/L_7)}{K_P(W_6/L_6)}}(V_{BIAS} - V_{SS} - V_{TN})$$



Output stage schematic

Example

If $W_7/L_7 = W_6/L_6$, $V_{DD} = 2.5V$, $V_{SS} = -2.5V$, and $V_{Bias} = 0V$, the trip point for the circuit shown is:

$$V_{TRP} = 2.5 - 0.7 - \sqrt{110/50}(0 + 2.5 - 0.7) = -0.870V$$

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Propagation Delay Time

- The propagation delay time is found from: $i_i = C_i \frac{dv_i}{dt_i} = C_i \frac{\Delta v_i}{\Delta t_i}$

where C_i is the capacitance to ground at the output of the i-th stage. ΔV is dependent on the previous state of the comparator.

$$t_i = \Delta t_i = C_i \frac{\Delta V_i}{I_i}$$

- The propagation delay time is found by summing the delays of each stage: $t_p = t_1 + t_2 + t_3 + \dots$



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Example: Propagation Delay Time (continued)

Solution

1. Total delay = sum of the first and second stage delays, t_1 and t_2 .
2. Consider the change of v_{G2} from -2.5V to 2.5V at 0.2 μ s.
3. $v_{o1} = +2.5$ V and $v_{out} = -2.5$ V.
4. t_{fo1} , requires C_I , I_5 , and ΔV_{o1} .
 - $C_I = 0.2$ pF
 - $I_5 = 30$ μ A
 - ΔV_1 can be calculated by finding the trip point of the output stage.



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Example: Propagation Delay Time (continued)

5. The trip point of the output stage by setting the current of M6 when saturated equal to $234 \mu\text{A}$:

$$\frac{\beta_6}{2}(V_{SG6} - |V_{TP}|)^2 = 234 \mu\text{A} \rightarrow v_{SG6} = 0.7 + \sqrt{\frac{234 \cdot 2}{110 \cdot 38}} = 1.035\text{V}$$

Therefore, the trip point of the second stage is:

$$V_{TRP2} = 2.5 - 1.035 = 1.465\text{V}$$

Therefore:

$$\Delta V_1 = 2.5\text{V} - 1.465\text{V} = V_{SG6} = 1.035\text{V}$$

Thus, the falling propagation time delay of the first stage is:

$$t_{fol} = 0.2\text{pF} \left(\frac{1.035\text{V}}{30\mu\text{A}} \right) = 6.9\text{ns}$$



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Example: Propagation Delay Time (continued)

6. The rising propagation delay time of the second stage requires C_{II} , ΔV_{out} , and I_6 . C_{II} is given as 5 pF, $\Delta V_{out} = 2.5V$ (assuming the trip point of the circuit connected to the output of the comparator is 0V), and I_6 can be found as follows:

$$V_{G6}(\text{guess}) \approx 0.5[V_{G6}(I_6 = 234\mu A) + V_{G6}(\text{min})]$$

$$V_{G6}(\text{min}) = V_{G1} - v_{g1}(I_{SS}/2) + V_{DS2} \approx -V_{GS1}(I_{SS}/2) = -0.7 - \sqrt{\frac{2 \cdot 15}{110 \cdot 3}} = -1.00V$$

$$V_{G6}(\text{guess}) \approx 0.5(1.465V - 1.00V) = 0.232V$$

Therefore, $V_{SG6} = 2.27V$ and

$$I_6 = \frac{\beta_6}{2}(V_{SG6} - |V_{TP}|)^2 = \frac{38 \cdot 50}{2}(2.27 - 0.7)^2 = 2,342\mu A$$



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Example: Propagation Delay Time (continued)

7. The rising propagation delay time for the output can expressed as:

$$t_{r_{out}} = 5\text{pF} \left(\frac{2.5\text{V}}{2,342\mu\text{A} - 234\mu\text{A}} \right) = 5.93\text{ns}$$

Thus, the total propagation delay time of the rising output of the comparator is approximately 12.8ns and most of this delay is attributable to the first stage.



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Example: Propagation Delay Time (continued)

8. Consider the change of v_{G2} from 2.5V to -2.5V, which occurs at 0.4 μ s. Therefore, $v_{o1} \approx V_{SS} = -2.5$ V and $v_{out} \approx V_{DD}$. The propagation delay times for the first and second stages are calculated as:

$$t_{r_{o1}} = 0.2\text{pF} \left(\frac{1.465\text{V} - (-1.13\text{V})}{30\mu\text{A}} \right) = 17.3\text{ns}$$

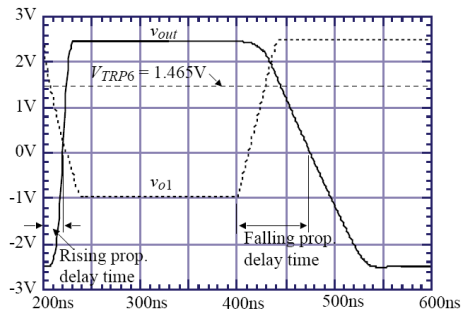
$$t_{f_{out}} = 5\text{pF} \left(\frac{2.5\text{V}}{234\mu\text{A}} \right) = 53.42\text{ns}$$



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Example: Propagation Delay Time (continued)

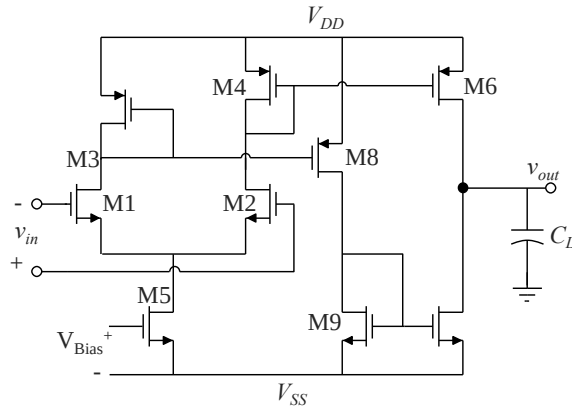
9. The total propagation delay time of the falling output is 70.72ns. Taking the average of the rising and falling propagation delay times gives a propagation delay time for this two-stage open-loop comparator of about 41.76ns.



Plot of propagation delay time

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Open-Loop Comparators: Push-Pull

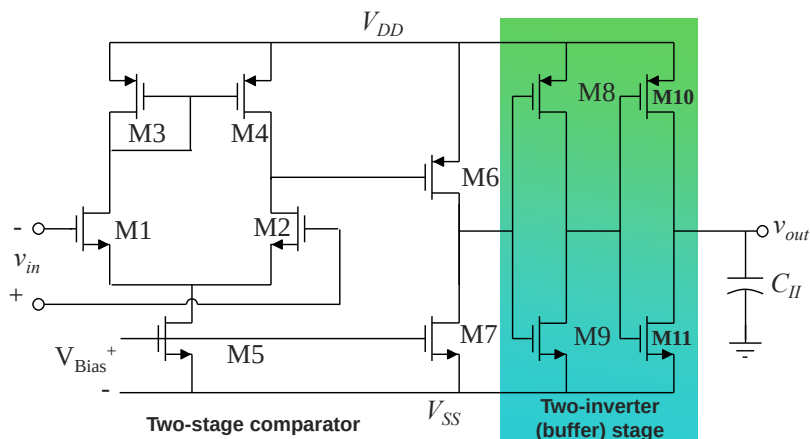


Push-pull comparator schematic

- Gain reduced $\rightarrow$ larger input resolution
- Push-pull output $\rightarrow$ balanced rise and fall slew rates

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Comparators That Drive Large Capacitive Loads



Typical specifications

- Slew rate = 3 V/ μ s into 50 pF
- Linear rise/fall time = 100 ns into 50 pF
- Propagation delay time $\approx$ 1 μ s (depending on bias currents and technology)
- Loop gain $\approx$ 32,000 V/V

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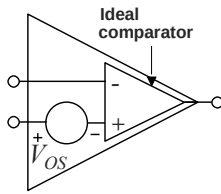
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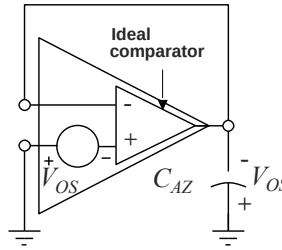
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Autozeroing Techniques

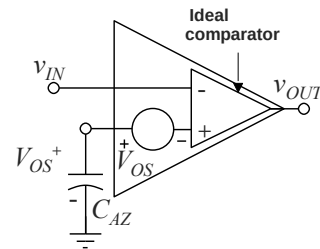
- Use the comparator as an opamp to sample the DC input offset voltage and cancel the offset during operation.



Comparator model



Autozero cycle



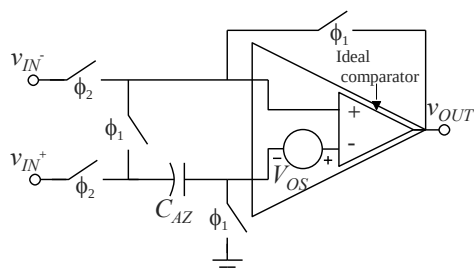
Comparison cycle

Disadvantages

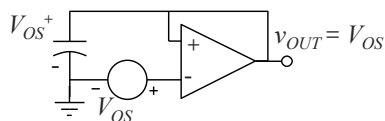
- The comparator must be stable in the unity-gain mode. (Self-compensating comparators are good; the two-stage op comparator requires compensation when switched during the autozero cycle.)
- Complete offset cancellation is limited by charge injection.

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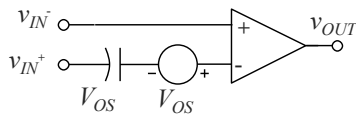
Differential Implementation: Autozeroed Comparators



Differential autozeroed comparator



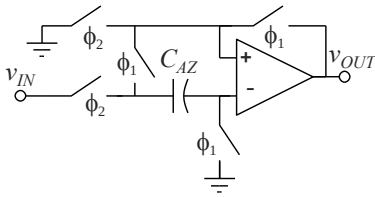
Comparator during ϕ_1 phase



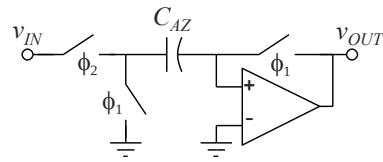
Comparator during ϕ_2 phase

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Autozeroed Comparator: Single-Ended Implementation



Noninverting configuration



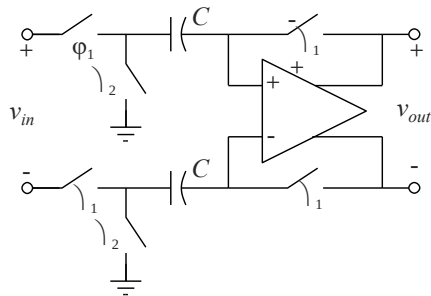
Inverting configuration

Note: Be careful about noise that gets sampled onto the autozeroing capacitor and is present on the comparison phase of the process.

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Discrete-Time Comparators

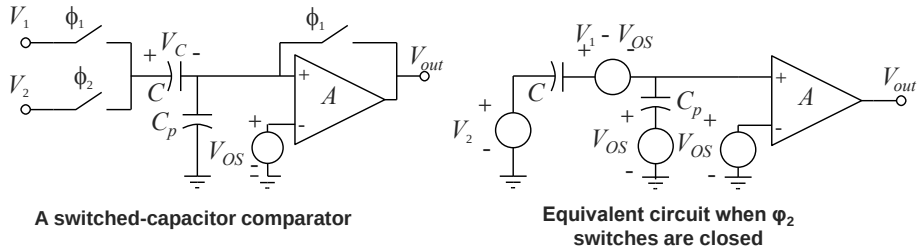
- Reduces the influence of charge injection
- Eliminates even harmonics



Differential-in, differential-out switched-capacitor comparator

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Switched-Capacitor Comparator



ϕ_1 phase: The V_1 input is sampled, and the DC input offset voltage is autozeroed.

$$V_C(\phi_1) = V_1 - V_{OS} \quad \text{and} \quad V_{C_p}(\phi_1) = V_{OS}$$

ϕ_2 phase:

$$\begin{aligned} V_{out}(\phi_2) &= -A \left[\frac{V_2 C}{C + C_p} - \frac{(V_1 - V_{OS})C}{C + C_p} - \frac{V_{OS} C_p}{C + C_p} \right] + A V_{OS} \\ &= -A \left[(V_2 - V_1) \frac{C}{C + C_p} + V_{OS} \left(\frac{C}{C + C_p} + \frac{C_p}{C + C_p} \right) \right] + A V_{OS} = -A(V_2 - V_1) \frac{C}{C + C_p} \approx A(V_2 - V_1) \end{aligned}$$

if C_p is smaller than C .

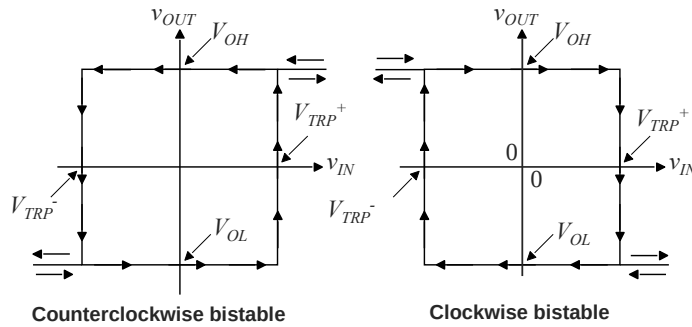
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With Hysteresis: Noisy Environments

Hysteresis is achieved by the use of positive feedback:

- Externally
- Internally

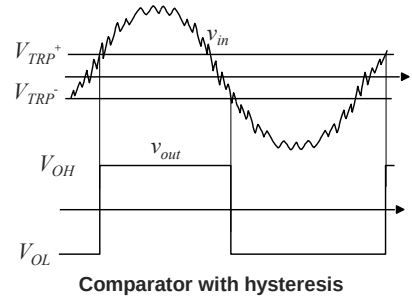
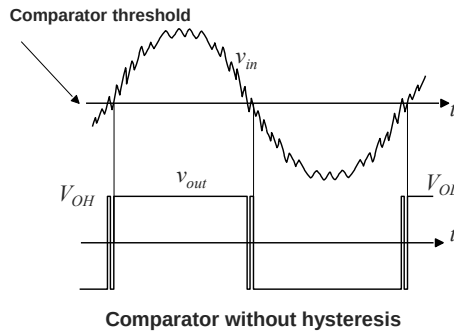
Transfer curve of a comparator with hysteresis



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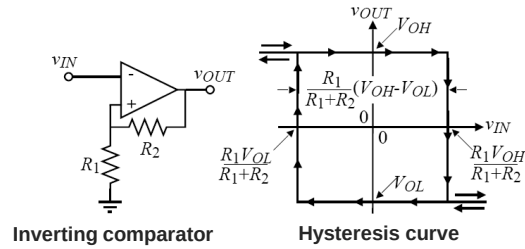
Input Noise With and Without Hysteresis

The figures show the influence of noise with and without hysteresis.



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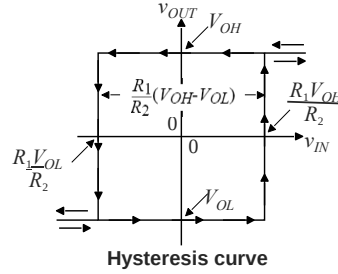
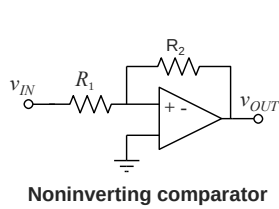
Inverting Comparator: External Positive Feedback



- **Upper trip point:** $v_{IN} = V_{TRP}^+ = \left(\frac{R_1}{R_1 + R_2} \right) V_{OH}$
- **Lower trip point:** $v_{IN} = V_{TRP}^- = \left(\frac{R_1}{R_1 + R_2} \right) V_{OL}$
- **Width of the bistable characteristic:** $\Delta V_{in} = V_{TRP}^+ - V_{TRP}^- = \left(\frac{R_1}{R_1 + R_2} \right) (V_{OH} - V_{OL})$

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Noninverting Comparator: External Positive Feedback



- **Upper trip point:** Assume that $v_{OUT} = V_{OL}$, the upper trip point occurs when

$$0 = \left(\frac{R_1}{R_1 + R_2} \right) V_{OL} + \left(\frac{R_2}{R_1 + R_2} \right) V_{TRP}^+ \rightarrow V_{TRP}^+ = -\frac{R_1}{R_2} V_{OL}$$

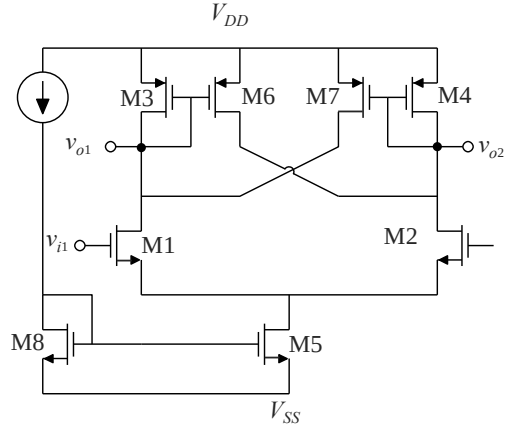
- **Lower trip point:** Assume that $v_{OUT} = V_{OH}$, the lower trip point occurs when

$$0 = \left(\frac{R_1}{R_1 + R_2} \right) V_{OH} + \left(\frac{R_2}{R_1 + R_2} \right) V_{TRP}^- \rightarrow V_{TRP}^- = -\frac{R_1}{R_2} V_{OH}$$

- **Width of the bistable characteristic:** $\Delta V_{in} = V_{TRP}^+ - V_{TRP}^- = \left(\frac{R_1}{R_2} \right) (V_{OH} - V_{OL})$

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Hysteresis: Internal Positive Feedback



Simple comparator with internal positive feedback

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Internal Positive Feedback: Upper Trip Point

Internal Positive Feedback - Upper Trip Point

Assume that the gate of M1 is on ground and the input to M2 is much smaller than zero. The resulting circuit is:

M1 on, M2 off $\rightarrow$ M3 and M6 on, M4 and M7 off.

$\therefore v_{o2}$ is high.

M6 would like to source the current $i_6 = \frac{W_6/L_6}{W_3/L_3} i_1$

As v_{in} begins to increase towards the trip point, the current flow through M2 increases. When $i_2 = i_6$, the upper trip point will occur.

$$\therefore i_5 = i_1 + i_2 = i_3 + i_6 = i_3 + \left(\frac{W_6/L_6}{W_3/L_3}\right) i_3 = i_3 \left[1 + \frac{W_6/L_6}{W_3/L_3}\right] \rightarrow i_1 = i_3 = \frac{i_5}{1 + [(W_6/L_6)/(W_3/L_3)]}$$

Also, $i_2 = i_5 - i_1 = i_5 - i_3$

Knowing i_1 and i_2 allows the calculation of v_{GS1} and v_{GS2} which gives

$$V_{TRP^+} = v_{GS2} - v_{GS1} = \sqrt{\frac{2i_2}{\beta_2}} + V_{T2} - \sqrt{\frac{2i_1}{\beta_1}} - V_{T1}$$

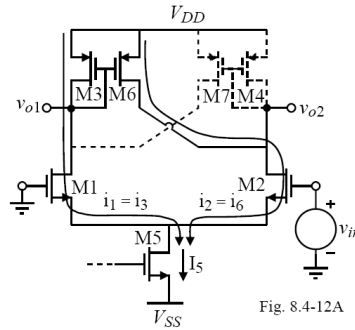


Fig. 8.4-12A

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Internal Positive Feedback: Lower Trip Point

Internal Positive Feedback - Lower Trip Point

Assume that the gate of M1 is on ground and the input to M2 is much greater than zero. The resulting circuit is:

M2 on, M1 off $\rightarrow$ M4 and M7 on, M3 and M6 off.

$\therefore v_{o1}$ is high.

M7 would like to source the current $i_7 = \frac{W_7/L_7}{W_4/L_4} i_2$

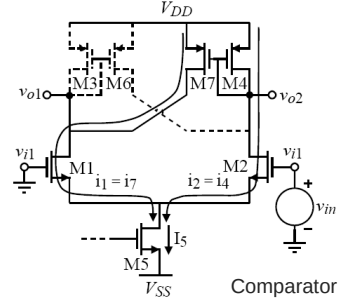
As v_{in} begins to decrease towards the trip point, the current flow through M1 increases. When $i_1 = i_7$, the lower trip point will occur.

$$\therefore i_5 = i_1 + i_2 = i_7 + i_4 = \left(\frac{W_7/L_7}{W_4/L_4} \right) i_4 + i_4 = i_4 \left[1 + \frac{W_7/L_7}{W_4/L_4} \right] \rightarrow i_2 = i_4 = \frac{i_5}{1 + [(W_7/L_7)/(W_4/L_4)]}$$

Also, $i_1 = i_5 - i_2 = i_5 - i_4$

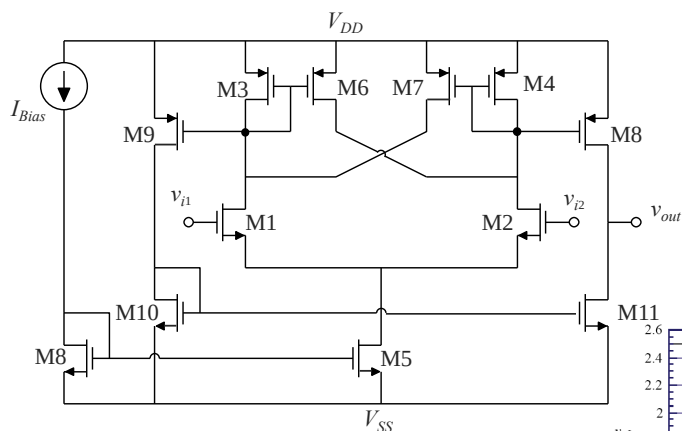
Knowing i_1 and i_2 allows the calculation of v_{GS1} and v_{GS2} which gives

$$V_{TRP} = v_{GS2} - v_{GS1} = \sqrt{\frac{2i_2}{\beta_2}} + V_{T2} - \sqrt{\frac{2i_1}{\beta_1}} - V_{T1}$$

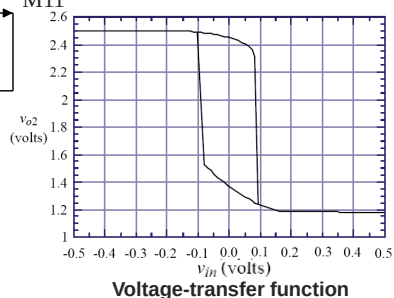


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Complete Circuit with Internal Positive Feedback



Comparator with internal positive feedback

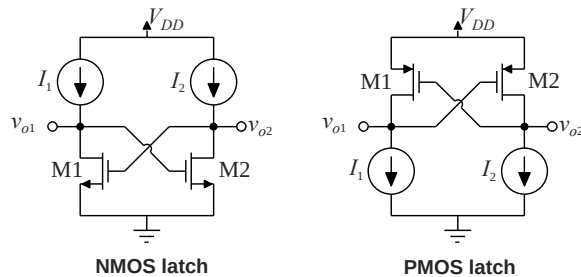


Voltage-transfer function

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Regenerative Comparators

- Regenerative comparators use positive feedback to accomplish the comparison of two signals. Latches have a faster switching speed than the previous bistable comparators.



- How is the input applied to a latch?
- The inputs are initially applied to the outputs of the latch:
 - $v_{o1'}$ = initial input applied to v_{o1}
 - $v_{o2'}$ = initial input applied to v_{o2}

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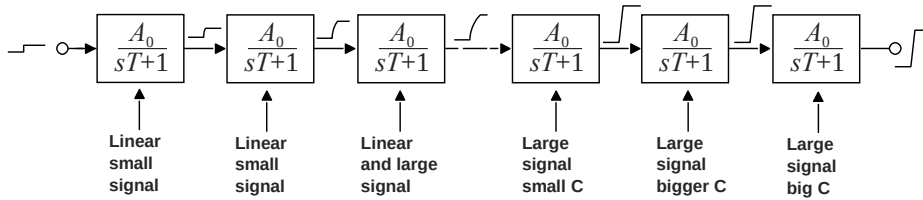
Regenerative Comparators (continued)

- Use positive feedback to get required gain instead of using multiple stages
- Use faster and lower power than previous open-loop designs



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High-Speed Comparators



Cascaded comparator concept

How does a cascaded, high-speed comparator work?

Assuming a small overdrive:

- The initial stage builds the driving capability.
- The latter stages swing rail to rail and build the ability to quickly charge and discharge capacitance.

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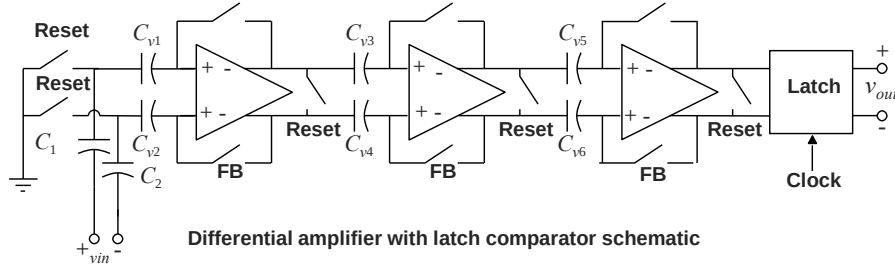
High-Speed Comparators (continued)

- Minimizing comparator propagation delay time:
 - The input signal is equal to $V_{in(min)}$ for worst case.
 - Amplifiers have a step response with a negative argument in the exponential.
 - Latches have a step response with a positive argument in the exponential.
- Use a cascade of linear amplifier to quickly build up the signal level and apply this amplified signal level to a latch for quick transition to the full binary output swing.



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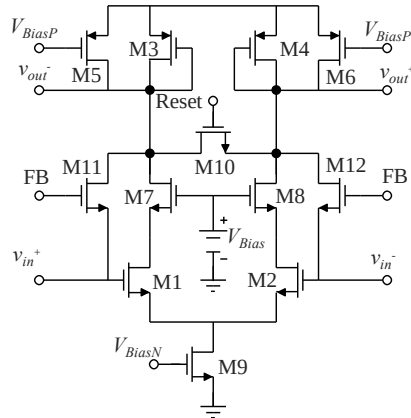
High-Speed Comparators (continued)



- Autozero and reset phase followed by comparison phase.
- More switches are needed to accomplish the reset and autozero of all preamplifiers simultaneously.
- Ability to run faster than 500 Msps.

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High-Speed Comparators (continued)



Example preamplifier circuit

$$\text{Gain} = A_v = -\frac{g_{m1}}{g_{m3}} = -\sqrt{\frac{K_N'(W_1/L_1)I_1}{K_P'(W_3/L_3)I_3}} = -\sqrt{\frac{K_N'(W_1/L_1)}{K_P'(W_3/L_3)}}\sqrt{\frac{I_5}{1+I_3}}$$

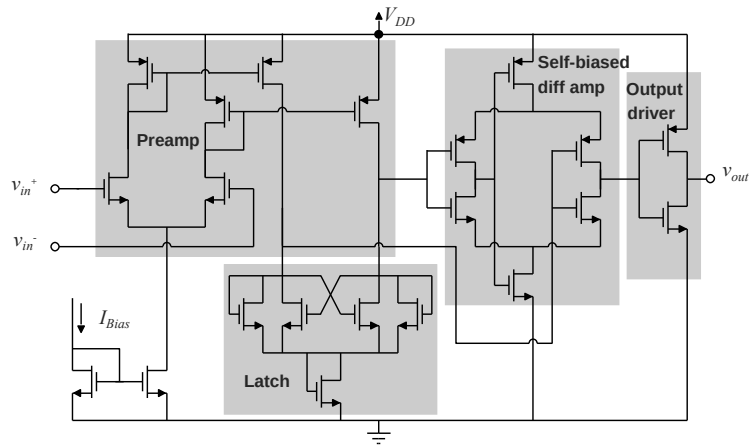
If $I_5 = 24I_3$, the gain is increased by a factor of 5

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High-Speed Comparators (continued)

Designed to have a $t_p = 10$ ns with a 5 pF load and a 10 mV overdrive,

- High-speed comparators are not synchronous.
- The comparator gain is greater than 2000 V/V, and the quiescent current is 100 μ A.



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Comparators Summary: Types

Types of comparators

- High gain, open loop
- Improved high-gain, open-loop comparators
 - Hysteresis
 - Autozeroing
- Regenerative comparators
- Discrete-time comparators



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Comparators Summary: Performance

Performance characterization

- Propagation delay time
- Binary output swing
- Input resolution and/or gain
- Input offset voltage
- Power dissipation



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Important Comparator Principles

- Regenerative comparators *should be optimized* to minimize the regeneration time constant.
- Capacitive imbalance in regenerative comparators *results in offset voltages*.
- The speed of the comparator depends on the linear and slewing responses, the overdrive, and the prior state.
- The DC input offset voltage depends on the matching and can be reduced by autozeroing.
- Charge injection is the limit of autozeroing.
- The open-loop comparator gain should be large enough for a binary output when $v_{in} = V_{in}(\min)$.
- In cascaded comparators, the first stages should have large GB and the last stages high SR.



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Introduction: Stability and Compensation

- Stability
 - Linear time-invariant systems
 - Nyquist criterion
 - Stability of amplifiers
 - Requirements on open-loop gain
 - Bode plots – gain and phase margin
- Compensation
 - Uncompensated frequency response of a two-stage opamp
 - Miller compensation
 - Left-half-plane (LHP) poles and right-half-plane (RHP) zero
 - Conditions for stability
 - Methods of controlling the RHP zero
 - Feedforward compensation
 - Self-compensation



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Linear Time-Invariant Systems: Stability

An LTI (linear time-invariant) system that is described by the transfer function (represented by $T(s)$ with M zeros and N poles) is stable if all poles have real negative parts. (All poles are situated in the LHP.)

The Bode criterion states that a feedback system is stable if the Bode plot of the loop gain magnitude of $A(s)b(s)$ is less than one when the phase crosses 180° . This means you can stabilize an unstable system by either:

- Adding feedback
- Moving poles and zeroes



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Where Do Poles Move to?

Need to be careful to distinguish among

- Open-loop gain $A(s)$
- Closed-loop gain $\frac{A(s)}{1 + A(s)\beta(s)}$
- Loop gain $A(s)\beta(s)$

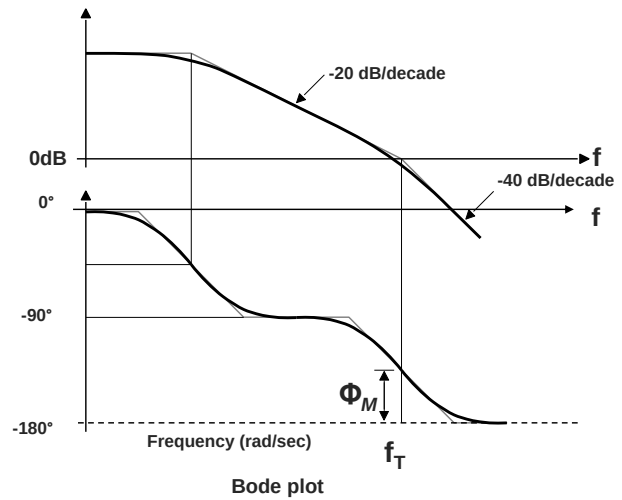
Where are poles of closed-loop transfer function?



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Stability Requirement: Bode Plots of Loop Gain

- A measure of stability is given by the difference between the critical -180° value and the phase at which $|A(f)\beta(f)| = 1$.
- This difference is called *phase margin*.
- *Negative* phase margin implies instability.

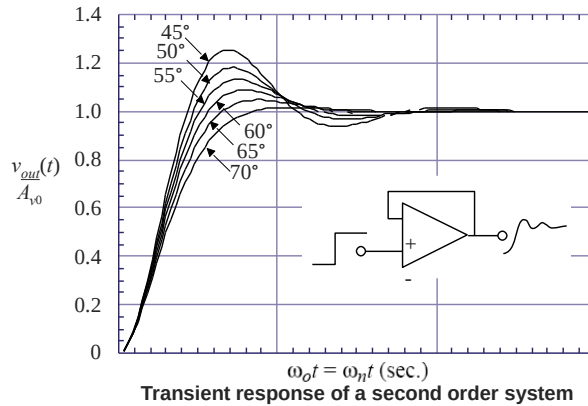


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Why Do You Want Good Stability?

Consider the step response of second-order system, which closely models the closed-loop gain of the opamp.

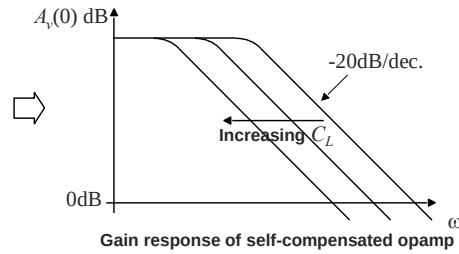
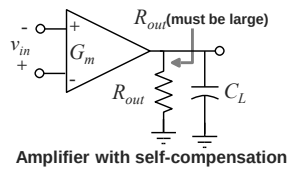
- A good step response is one that quickly reaches its final value. Therefore, you can see that phase margin should be at least 45° and preferably 60° or larger. A standard guideline for satisfactory stability is that there should be less than three rings.
- A second good reason for good stability is that due to production tolerances a near-stable amplifier can turn into an oscillator, which kills the circuit function completely.



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Self-Compensated Opamps

Self-compensation occurs when the load capacitor is the compensation capacitor. (It can never be unstable for resistive feedback.)



Voltage gain: $\frac{v_{out}}{v_{in}} = A_v(0) = G_m R_{out}$

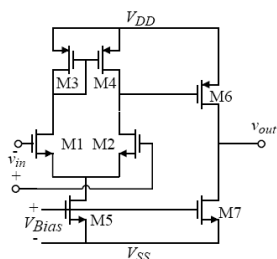
Dominant pole: $p_1 = \frac{-1}{R_{out} C_L}$

Unity-gain bandwidth: $GB = A_v(0) \cdot |p_1| = \frac{G_m}{C_L}$

Stability: Large load capacitors reduce GB, but the phase is still 90° at GB.

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Two-Stage Opamp Uncompensated Frequency Response



Two-stage opamp schematic

$$\begin{aligned}g_{m1} &= 1 \text{ mA/V} \\C_{gs6} &= 1 \text{ pF} \\r_{o1} &= 160 \text{ k}\Omega\end{aligned}$$

$$p_1 = 1 \text{ MHz}$$

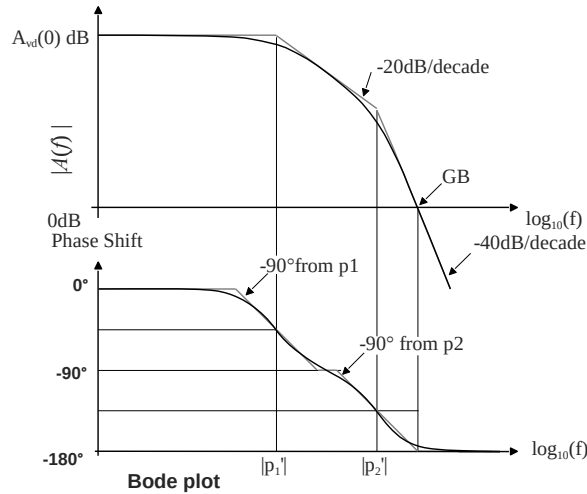
$$\begin{aligned}g_{m6} &= 10 \text{ mA/V} \\C_L &= 10 \text{ pF} \\r_{out} &= 5 \text{ k}\Omega\end{aligned}$$

$$p_2 = 3 \text{ MHz}$$

Note: C_{gs6} , C_L used in this example to model two lowest frequency poles. Mirror pole from M3 is at a much higher frequency due to low impedance of diode-connected M3.

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Opamp Uncompensated Frequency Response



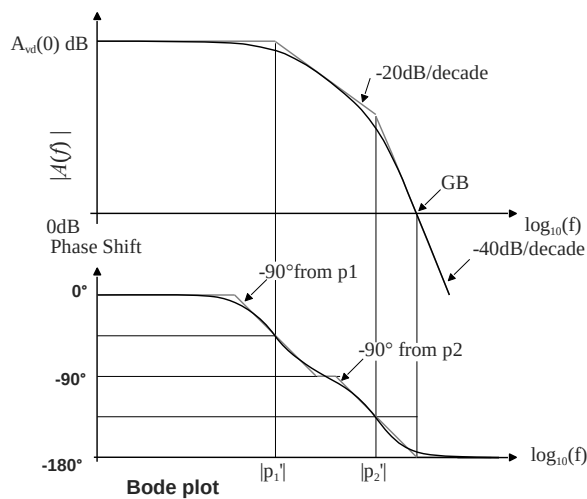
- If you assume that $\beta(s) = 1$ (this is the worst case for stability considerations and is often used to form a unity-gain buffer), then the plot is the same as the loop gain.
- Notice that the phase margin is much less than 45°. Therefore, the opamp must be compensated before using it in a closed-loop configuration (at least for a unity-gain buffer).

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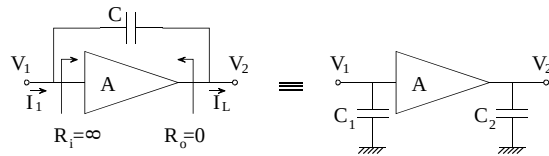
Compensating Opamp: Move p1



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Miller Effect

A capacitor present in the feedback path of an ideal voltage amplifier is equivalent to two capacitors loading the input and output of the amplifier. This is called the *Miller Effect*.



Ideal amplifier with capacitive feedback and Miller equivalent

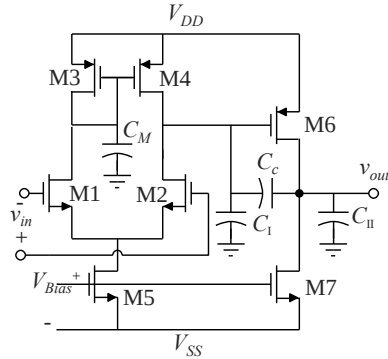
The formulas for C_1 and C_2 are:

$$C_1 = (1 - A)C$$

$$C_2 = \frac{A - 1}{A}C$$

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Miller Compensation



Two-stage opamp

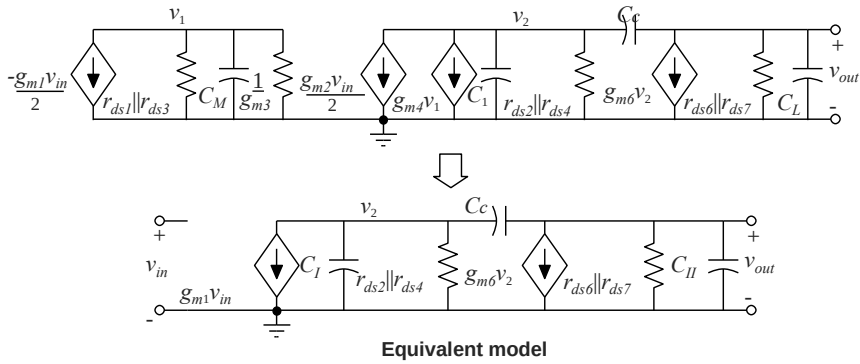
The various capacitors are:

- C_c = accomplishes the Miller compensation
- C_M = capacitance associated with the first-stage mirror (mirror pole)
- C_I = output capacitance to ground of the first-stage
- C_{II} = output capacitance to ground of the second-stage

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Miller Small-Signal Frequency-Response Model

1. Assume that $g_{m3} \gg g_{ds3} + g_{ds1}$
2. Assume that $g_{m3} / C_M \gg GB$



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Miller Compensation: Two-Stage Opamps

Three roots are important:

- RHP zero: $z_1 = \frac{g_{mII}}{C_c} = \frac{g_{m6}}{C_c}$

This root is very undesirable. It boosts the magnitude while decreasing the phase.

- Dominant LHP pole (the Miller pole):

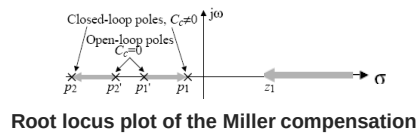
$$p_1 \approx \frac{-1}{g_{mII} R_I R_{II} C_c} = \frac{-(g_{ds2} + g_{ds4})(g_{ds6} + g_{ds7})}{g_{m6} C_c}$$

This root accomplishes the desired compensation.

- LHP output pole:

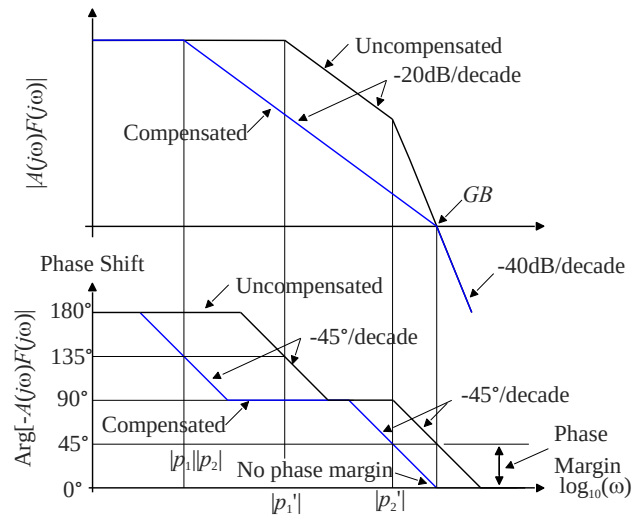
$$p_2 \approx \frac{-g_{mII}}{C_{II}} \approx \frac{-g_{m6}}{C_L}$$

(This pole must be $\geq$ the unity-gain bandwidth or the phase margin is not be satisfied.)



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Compensated Open-Loop Frequency Response



$$GB = A_{vd}(0) \cdot |p_1| = (g_{mI} g_{mII} R_I R_{II}) \frac{1}{g_{mII} R_I R_{II} C_c} = \frac{g_{mI}}{C_c} = \frac{g_{m1}}{C_c} = \frac{g_{m2}}{C_c}$$

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Where Do Poles and Zeroes Come From?

- The Miller pole:

$$|p_1| \approx \frac{1}{R_I(g_{m6}R_{II}C_c)}$$

- The LHP output pole:

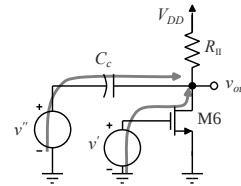
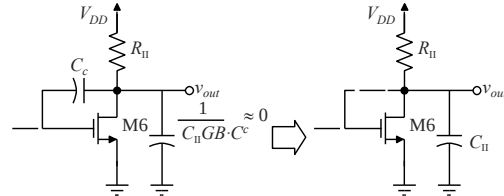
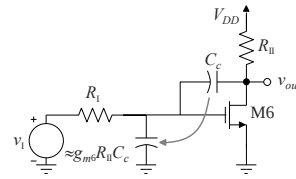
$$|p_2| \approx \frac{g_{m6}}{C_{II}}$$

- RHP zero:

Note: Zeros always arise from multiple paths from the input to output.

$$v_{out} = \left(\frac{-g_{m6}R_{II}(1/sC_c)}{R_{II} + 1/sC_c} \right) v' + \left(\frac{R_{II}}{R_{II} + 1/sC_c} \right) v'' = \frac{-R_{II} \left(\frac{g_{m6}}{sC_c} - 1 \right)}{R_{II} + 1/sC_c} v$$

where $v = v' = v''$.



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Conditions for Stability of the Two-Stage Opamp

Assume $p \geq GB$

- Unity-gain bandwidth is given as: $GB = A_v(0) \cdot |p_1| = \left(\frac{g_{m1} g_{m2} R_I R_{II}}{g_{m1} R_I R_{II} C_c} \right) = \frac{g_{m1}}{C_c} = \frac{g_{m1} g_{m2} R_I R_{II}}{g_{m2} R_I R_{II} C_c} = \frac{g_{m1}}{C_c}$
- The requirement for 45° phase margin is:

$$\pm 180^\circ - \text{Arg}[AF] = \pm 180^\circ - \tan^{-1}\left(\frac{\omega}{|p_1|}\right) - \tan^{-1}\left(\frac{\omega}{|p_2|}\right) - \tan^{-1}\left(\frac{\omega}{z}\right) = 45^\circ$$

- Letting $\omega = GB$ and assuming that $z \geq 10GB$ gives:

$$\pm 180^\circ - \tan^{-1}\left(\frac{GB}{|p_1|}\right) - \tan^{-1}\left(\frac{GB}{|p_2|}\right) - \tan^{-1}\left(\frac{GB}{z}\right) = 45^\circ$$

$$135^\circ \approx \tan^{-1}(A_v(0)) + \tan^{-1}\left(\frac{GB}{|p_2|}\right) + \tan^{-1}(0.1) = 90^\circ + \tan^{-1}\left(\frac{GB}{|p_2|}\right) + 5.7^\circ$$

$$39.3^\circ \approx \tan^{-1}\left(\frac{GB}{|p_2|}\right) \Rightarrow \frac{GB}{|p_2|} = 0.818 \Rightarrow |p_2| \geq 1.22GB$$

- The requirement for 60° phase margin is:

$$|p_2| \geq 2.2GB \text{ if } z \geq 10GB$$

- If 60° phase margin is required, then the following relationships apply:

$$\frac{g_{m6}}{C_c} > \frac{10 g_{m1}}{C_c} \Rightarrow g_{m6} > 10 g_{m1} \text{ and } \frac{g_{m6}}{C_2} > \frac{2.2 g_{m1}}{C_c} \Rightarrow C_c > 0.22 C_2$$

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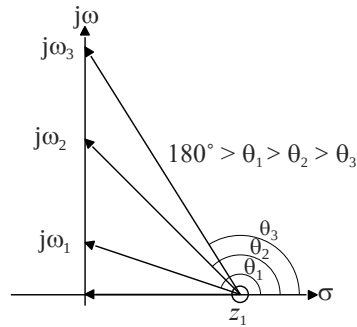


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Controlling the RHP Zero

Question: Why is the RHP zero a problem?

Answer: Because it boosts the magnitude but lags the phase, which is the worst possible combination for stability.

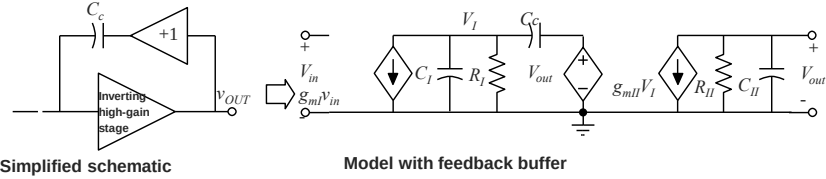


Solution: If zeros are caused by two paths to the output, then eliminate one of the paths.

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Eliminating Feedforward Path with Buffer

Use of a buffer to eliminate feedforward path through the Miller capacitor model:



The transfer function is given by the following equation:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{g_{mI} g_{mII} R_I R_{II}}{1 + s[R_I C_I + R_{II} C_{II} + R_I C_c + g_{mII} R_I R_{II} C_c] + s^2[R_I R_{II} C_{II} (C_I + C_c)]}$$

Using the technique as before to approximate p_1 and p_2 results in the following:

$$p_1 \cong \frac{-1}{R_I C_I + R_{II} C_{II} + R_I C_c + g_{mII} R_I R_{II} C_c} \cong \frac{-1}{g_{mII} R_I R_{II} C_c} \quad \text{and} \quad p_2 = \frac{-g_{mII} C_c}{R_I R_{II} C_{II} (C_I + C_c)}$$

Poles are approximately what they were before with the zero removed:

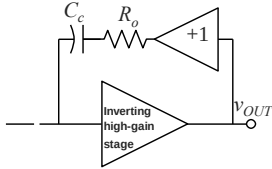
- For 45° phase margin, $|p_2|$ must be greater than GB.
- For 60° phase margin, $|p_2|$ must be greater than 1.73 GB.



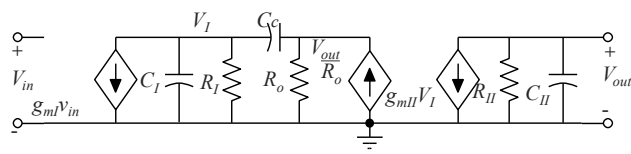
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Eliminating RHP Zero with Buffer

Use a buffer with finite output resistance to eliminate the RHP zero. Assume that the unity-gain buffer has an output resistance of R_o .



Simplified schematic



Model with feedback buffer and resistor

It can be shown that if the output resistance of the buffer amplifier, R_o , is not neglected then another pole occurs at:

$$p_4 \cong \frac{-1}{R_o [C_I C_c / (C_I + C_c)]}$$

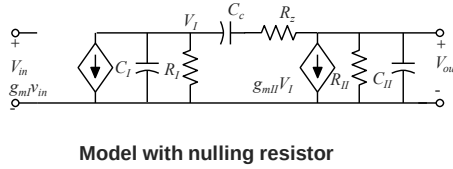
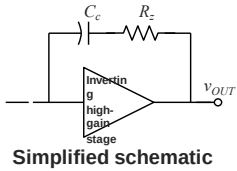
and a LHP zero at: $z_2 \cong \frac{-1}{R_o C_c}$

Closer examination shows that if a resistor, called a *nulling resistor*, is placed in series with C_c that the RHP zero can be eliminated or moved to the LHP.

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Eliminating RHP Zero with Nulling Resistor

- Use a nulling resistor to eliminate the RHP zero (or turn it into an LHP zero).



- Nodal equations

$$g_{ml} V_{in} + \frac{V_I}{R_I} + sC_I V_I + \left(\frac{sC_c}{1 + sC_c R_z} \right) (V_I - V_{out}) = 0$$

$$g_{mII} V_I + \frac{V_o}{R_{II}} + sC_{II} V_{out} + \left(\frac{sC_c}{1 + sC_c R_z} \right) (V_{out} - V_I) = 0$$



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Eliminating RHP Zero with Nulling Resistor (continued)

Solution

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{a\{1 - s[(C_c / g_{mII}) - R_z C_c]\}}{1 + bs + cs^2 + ds^3}$$

Where:

$$a = g_{mI} g_{mII} R_I R_{II}$$

$$b = (C_{II} + C_c)R_{II} + (C_I + C_c)R_I + g_{mII}R_I R_{II} C_c + R_z C_c$$

$$c = [R_I R_{II} (C_I C_{II} + C_c C_I + C_c C_{II}) + R_z C_c (R_I C_I + R_{II} C_{II})]$$

$$d = R_I R_{II} R_z C_I C_{II} C_c$$



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Eliminating RHP Zero with Nulling Resistor (continued)

If R_z is assumed to be less than R_I or R_{II} , and the poles widely spaced, then the roots of the transfer function (from previous slide) can be approximated as:

$$p_1 \cong \frac{-1}{(1 + g_{mII} R_{II}) R_I C_c} \cong \frac{-1}{g_{mII} R_I R_{II} C_c}$$

$$p_2 \cong \frac{-g_{mII} C_c}{C_I C_{II} + C_c C_I + C_c C_{II}} \cong \frac{-g_{mII}}{C_{II}}$$

$$p_4 = \frac{-1}{R_z C_I}$$

and

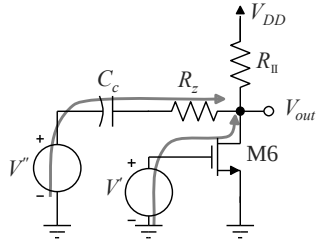
$$z_1 = \frac{1}{C_c (1/g_{mII} - R_z)}$$

Note: The zero can be placed anywhere on the real axis.



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Concept: Nulling Resistor Approach



Conceptual schematic of nulling resistor approach

The output voltage, V_{out} , can be written as:

$$V_{out} = \frac{-g_{m6}R_{II}\left(R_z + \frac{1}{sC_c}\right)}{R_{II} + R_z + \frac{1}{sC_c}}V' + \frac{R_{II}}{R_{II} + R_z + \frac{1}{sC_c}}V'' = \frac{-R_{II}\left[g_{m6}R_z + \frac{g_{m6}}{sC_c} - 1\right]}{R_{II} + R_z + \frac{1}{sC_c}}V$$

When $V = V' = V''$

Setting the numerator equal to zero and assuming $g_{m6} = g_{mII}$ gives:

$$z_1 = \frac{1}{C_c(1/g_{mII} - R_z)}$$

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Cancelling Output Pole with RHP Zero

To allow the RHP zero to cancel the output pole, p_2 , move the zero to the LHP:

You want $z_1 = p_2$ in terms of the previous notation.

Therefore, $\frac{1}{C_c(1/g_{mII} - R_z)} = \frac{-g_{mII}}{C_{II}}$

The value of R_z can be found as:

$$R_z = \left(\frac{C_c + C_{II}}{C_c} \right) (1/g_{mII})$$

With p_2 canceled, the remaining roots are p_1 and p_4 (the pole due to R_z). For unity-gain stability, all that is required is that:

$$|p_4| > A_v(0)|p_1| = \frac{A_v(0)}{g_{mII} R_{II} R_I C_c} = \frac{g_{mI}}{C_c}$$

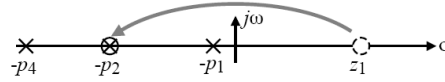
and

$$(1/R_z C_I) > (g_{mI}/C_c) = GB$$

Substituting R_z into the above inequality and assuming $C_{II} \gg C_c$ results in:

$$C_c > \sqrt{\frac{g_{mI}}{g_{mII}}} C_I C_{II}$$

This procedure gives excellent stability for a fixed value of C_{II} ($\approx C_L$). Unfortunately, as C_L changes, p_2 changes and the zero must be readjusted to cancel p_2 .

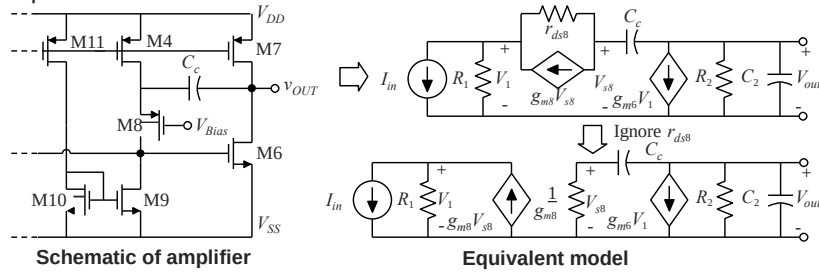


Cancelling the output pole using the zero

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Increasing Magnitude of Output Pole

The magnitude of the output pole, p_2 , can be increased by adding a current buffer in the Miller capacitor feedback path. The following amplifier has a current buffer and Miller compensation:



The resistors R_1 and R_2 are defined as:

$$R_1 = \frac{1}{g_{ds2} + g_{ds4} + g_{ds9}} \quad \text{and} \quad R_2 = \frac{1}{g_{ds6} + g_{ds7}}$$

where transistors M2 and M4 are the output transistors of the first stage.

Nodal equations

$$I_{in} = G_1 V_1 - g_{m8} V_{s8} = G_1 V_1 - \left(\frac{g_{m8} s C_c}{g_{m8} + s C_c} \right) V_{out} \quad \text{and} \quad 0 = g_{m6} V_1 + \left[G_2 + s C_2 + \frac{g_{m8} s C_c}{g_{m8} + s C_c} \right] V_{out}$$

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Increasing Magnitude of Output Pole (continued)

Solving for the transfer function V_{out}/I_{in} gives

$$\frac{V_{out}}{V_{in}} = \left(\frac{-g_{m6}}{G_1 G_2} \right) \left[\frac{\left(1 + \frac{sC_c}{g_{m8}} \right)}{1 + s \left[\frac{C_c}{g_{m8}} + \frac{C_2}{G_2} + \frac{C_c}{G_2} + \frac{g_{m6}C_c}{G_1 G_2} \right] + s^2 \left(\frac{C_c C_2}{g_{m8} G_2} \right)} \right]$$

Using the approximate method of solving for the roots of the denominator gives:

$$p_1 = \frac{-1}{\frac{C_c}{g_{m8}} + \frac{C_2}{G_2} + \frac{C_c}{G_2} + \frac{g_{m6}C_c}{G_1 G_2}} \approx \frac{-6}{g_{m6} r_{ds}^2 C_c}$$

and

$$p_2 \approx \frac{-\frac{g_{m6} r_{ds}^2 C_c}{6}}{\frac{C_c C_2}{g_{m8} G_2}} = \frac{g_{m8} r_{ds}^2 G_2}{6} \left(\frac{g_{m6}}{C_2} \right) |p_2'|$$

where all the various channel resistance have been assumed to equal r_{ds} and p_2' is the output pole for normal Miller compensation.

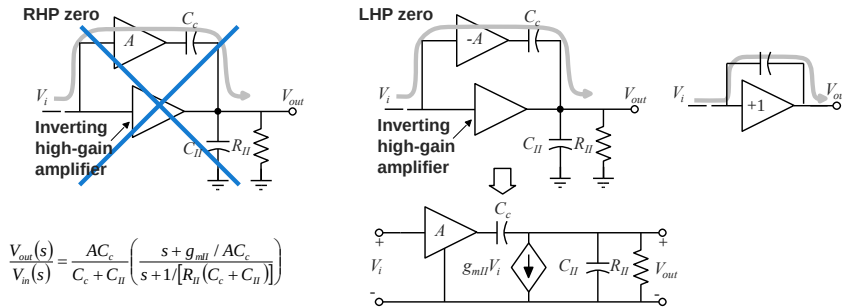
Result: Dominant pole is approximately the same and the output pole is increased by $\approx g_{m6} r_{ds}$.



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Feedforward Compensation

Use two parallel paths to achieve a LHP zero for lead compensation purposes.



To use the LHP zero for compensation, a compromise must be reached:

- Placing the zero *below* GB leads to boosting the loop gain, which can deteriorate the phase margin.
- Placing the zero *above* GB has less influence on the leading phase caused by the zero.
- A source follower is a good candidate for the use of feedforward compensation.

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Summary: Stability and Compensation

- Stability
 - Linear time-invariant systems
 - Stability of amplifiers
 - Requirements on open-loop gain
 - Bode plots – gain and phase margin
- Compensation
 - Uncompensated frequency response of a two-stage opamp
 - Self-compensation
 - Miller compensation
 - Left-half-plane (LHP) poles and right-half-plane (RHP) zero
 - Conditions for stability
 - Methods of controlling the RHP zero
 - Feedforward compensation



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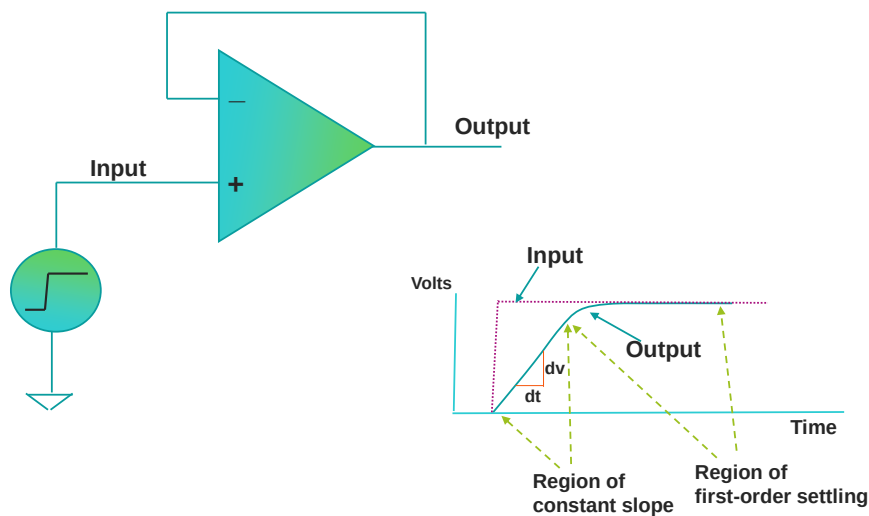
Slew Rate

- Many feedback loops exhibit settling behavior which is limited by the slew rate of the amplifier in the loop.
- When a system responds to a step response with a constant slope (dv/dt) at its output, the system is slew-rate limited.
- Amplifiers are often slew-rate limited when a maximum current is limited by fixed current sources in the amplifier.
 - For 2-stage op-amps, may be either input stage or output stage.
 - If input stage: Slew rate limiting usually means that input range of input transconductance stage is exceeded.
- When a slew-rate-limited system sees a step input, the first part of the settling time is a constant slope while the second section of the waveform reverts to a first- or second-order settling characteristic.



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Example of Slew-Rate-Limited Settling



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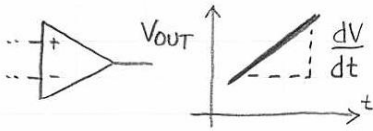


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Analysis of Slew-Rate-Limited Settling

SLEW RATE LIMITING

WHAT IT IS:



MAXIMUM dV/dt POSSIBLE
AT OP-AMP OUTPUT

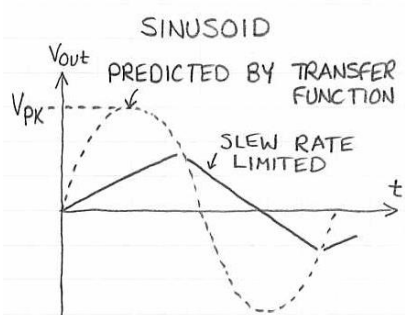
"SPEED LIMIT"

ANALYSIS PROCEDURE

- 1) DETERMINE OUTPUT $V_{out}(t)$
FROM LINEAR BEHAVIOR
(TRANSFER FUNCTION,
1ST ORDER STEP RESPONSE)
- 2) CALCULATE $dV_{out}(t)/dt$
- 3) SEE IF MAX $dV_{out}(t)/dt$
EXCEEDS SLEW RATE LIMIT

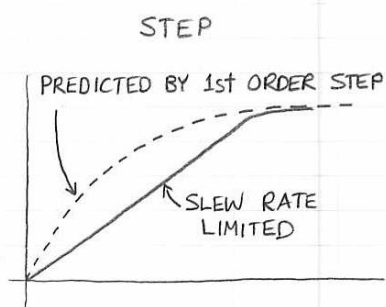
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Analysis of Slew-Rate-Limited Settling



$$V_{out}(t) = V_{PK} \sin(2\pi f t)$$

$$\frac{dV_{out}(t)}{dt} = \underbrace{2\pi f V_{PK}}_{\text{MAX } \frac{dV}{dt}} \underbrace{\cos(2\pi f t)}_{\text{MAX}=1}$$



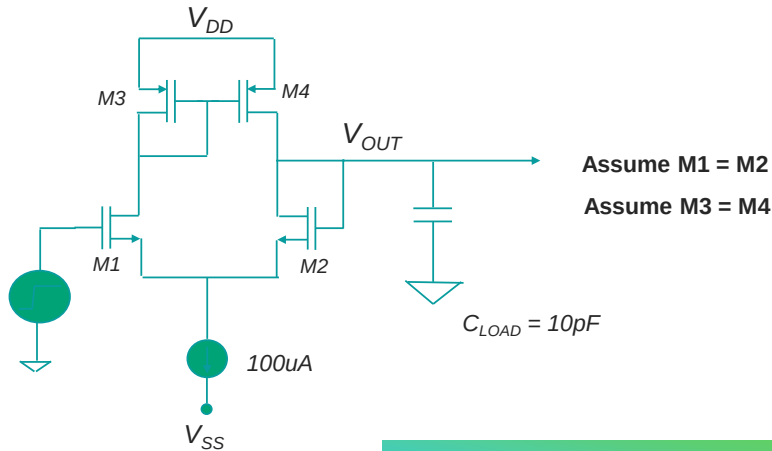
$$V_{out}(t) = V_F - (V_F - V_I) e^{-t/\tau}$$

$$\frac{dV_{out}(t)}{dt} = \underbrace{\frac{(V_F - V_I)}{\tau}}_{\text{MAX } \frac{dV}{dt}} \underbrace{e^{-t/\tau}}_{\text{MAX}=1 \text{ at } t=0}$$

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Slew Rate Example

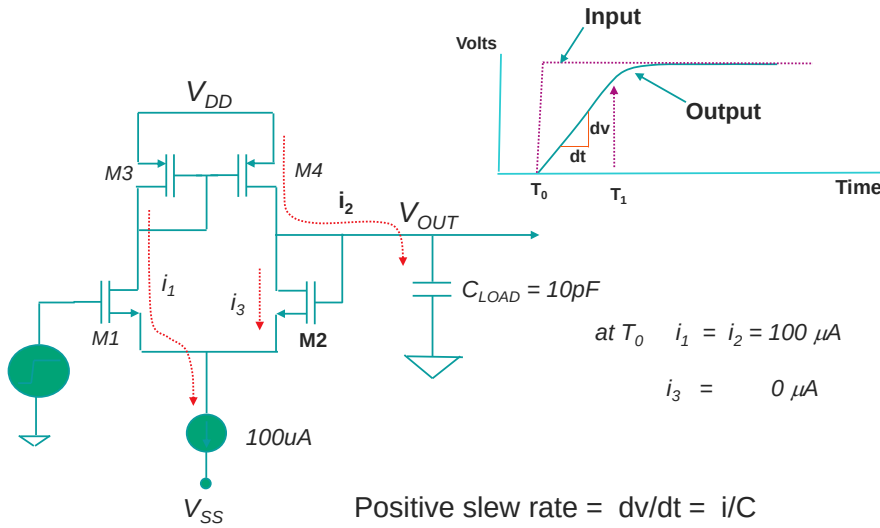
What is the maximum rising and falling slew rate of the buffer amplifier shown?



Note: This circuit is useful for generating pulses with limited, well-defined slew rates.

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Positive Slew Rate Calculation

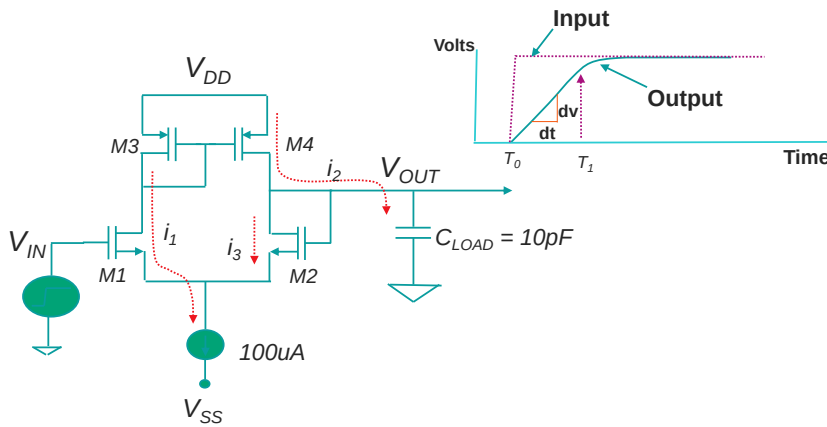


$$\text{Positive slew rate} = dv/dt = i/C$$
$$\text{Positive slew rate} = 100 \mu A / 10e^{-12} = 10e^6 \text{ V/sec}$$

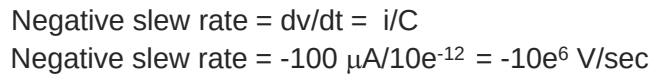
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Change from Slewing to Settling

At T_1 , V_{OUT} has come close enough to V_{IN} that M2 begins to conduct and the amplifier begins to exit slewing and starts to transition to a first-order settling characteristic:



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81

Real-World Settling

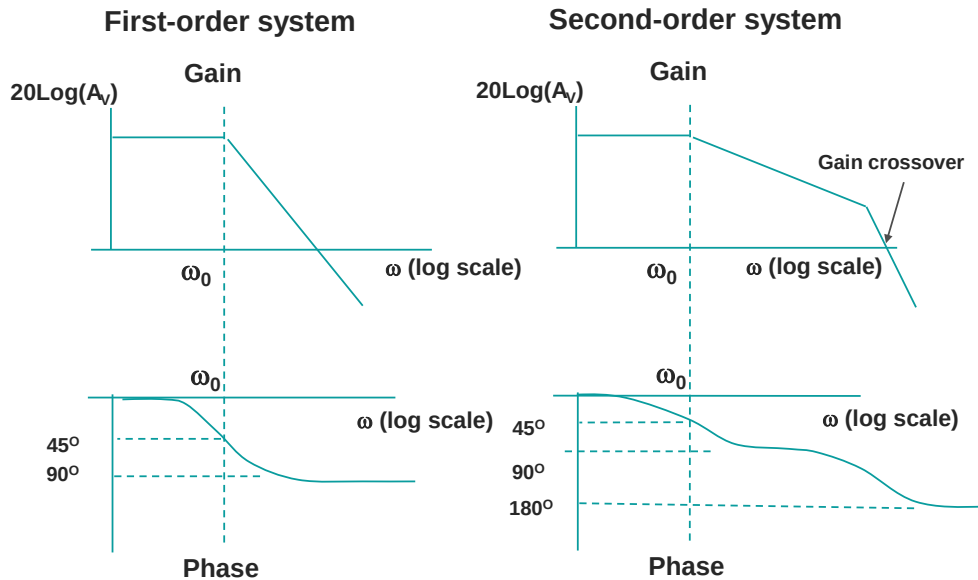
Usually, a combination of slew limited plus linear settling

- Slew rate limited
 - Rising, falling rates can be different
 - Increasing slew rate to avoid slew rate limited settling usually costs power
- Linear settling
 - Depends on phase margin
 - Can be first-order or second-order, or other order for more complicated opamps



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First-Order vs. Second-Order Gain or Phase Plots



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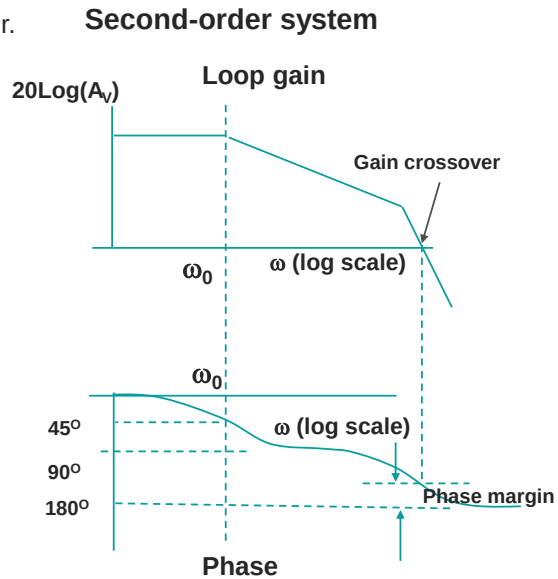


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Phase Margin

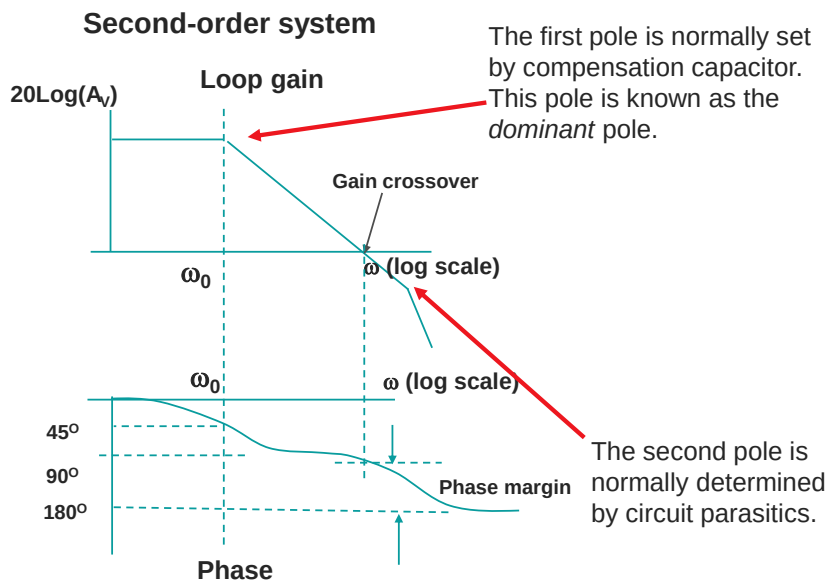
1. Find phase shift around loop at gain crossover.
2. Subtract this from 180° .

This difference is known as the *phase margin*. If phase margin is 0° or negative, the circuit oscillates.



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Design for Fast Settling



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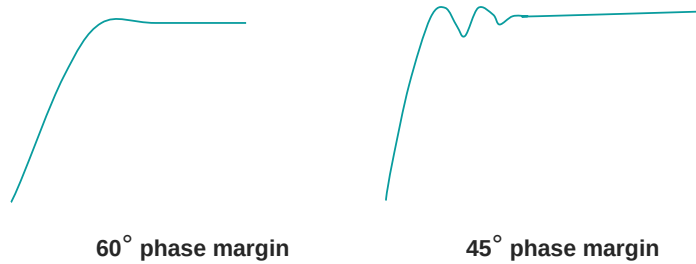


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Second-Order Settling

Try to push the second-pole break frequency to the right so that second-gain break point happens beyond gain crossover frequency:

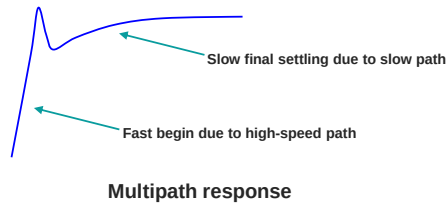
- Target $\sim 55 - 65^\circ$ phase margin for fast settling for most applications.
- More phase margin often means you are wasting power.
- A phase margin below 50° runs the risk of poor settling or oscillation.



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Other Settling Behaviors

Most systems behave as described so far, but there are also exceptions. For example, feedforward amplifiers can look slightly different. Those have a fast signal path and a slow one in parallel, which leads to such a response:



Note: Phase margin might be very good (for example, 85°), but settling is slower than BW indicates!

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Summary: Settling Behavior

- Settling behavior appears in first-order and higher systems, though the concept of types of settling requires second-order and higher:
 - Critically damped
 - Overdamped
 - Underdamped
- Settling is the time-domain counterpart to phase margin.
- Slew rate is also related to the settling behavior.
- Some systems exhibit a settling behavior that consist of a multipath response.



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Introduction: Tuned Amplifiers

Basic principle

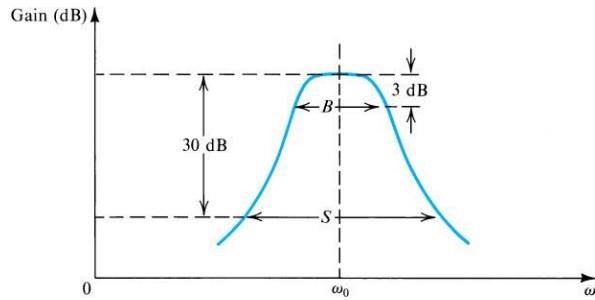
- Uses RLC circuit as load or input of an amplifier
- Designed to have a narrow-band response

Characteristics

- Resonant frequency
- Midband gain
- Bandwidth
- Quality factor

Applications

- Low-noise amplifiers
- IF amplifiers

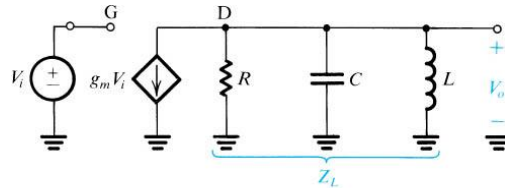
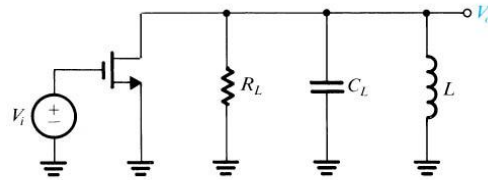


Frequency response of a tuned amplifier

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Basic Tuned Amplifier

- Voltage gain
$$V_o/V_i = \frac{-g_m s}{C(s^2 + s/RC + 1/LC)}$$
- Resonant frequency
$$\omega_o = 1/\sqrt{LC}$$
- 3-dB bandwidth
$$B = 1/RC$$
- Quality factor
$$Q = \omega_o / B = \omega_o RC$$
- Midband gain
$$V_o/V_i = -g_m R$$

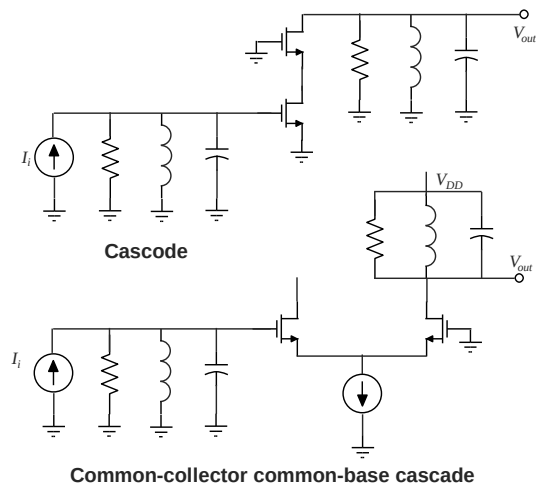


Single-tuned amplifier with parallel RLC load

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Multiple-Tuned Amplifiers

- Tuned amplifier configurations with two devices that do not suffer from the Miller effect are:
 - Cascode
 - Common-collector common-base cascade
- Multiple-tuned cascode amplifier:
 - Uses tuning circuit at both input and output.
 - Eliminates Miller effect by isolating input and output ports.
 - Provides greater selectivity than single-tuned amplifier.



Note: Bias details of the cascode circuit are not shown.

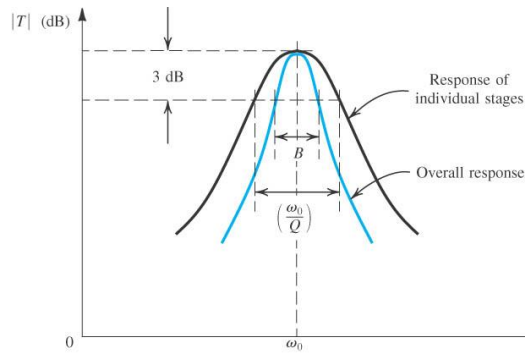
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Synchronously Tuned Amplifiers

Basic principle

- Use cascaded stages with identical resonant circuits.
- Overall response has a narrower bandwidth than individual stages.

- For n stages $B = \frac{\omega_0 \sqrt{2^{1/N} - 1}}{Q}$



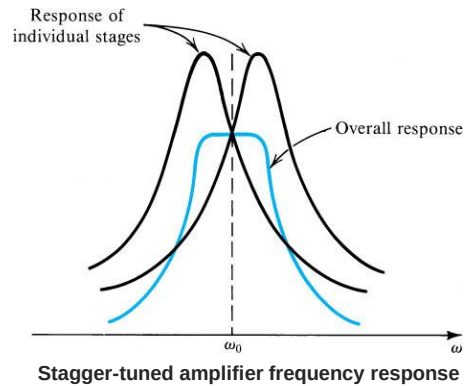
Frequency response of a synchronously tuned amplifier

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Stagger-Tuned Amplifiers

Basic principle

- Uses cascaded stages with resonant circuits having staggered resonant frequencies.
- Can result (stagger-tuning the individual resonant circuits) in an overall response with a passband flatter than that obtained with synchronous tuning.



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Summary: Tuned Amplifiers

- Tuned amplifiers allow you to center and set the bandwidth of the response.
- Several types of tuned amplifiers exist:
 - Single tuned
 - Multiple tuned
 - Synchronously tuned
 - Stagger tuned



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Introduction: Common-Mode Response

Differential amplifiers have many advantages over single-ended amplifiers:

- Larger output swing results in more dynamic range for a given supply
- High power-supply rejection rate
- Even-order harmonics rejected

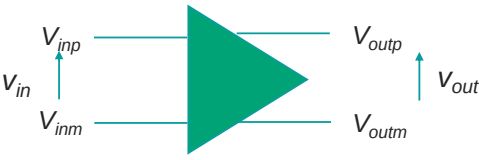
However, differential amplifiers require some method to set the output common-mode voltage.



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Output Common Mode

The output of a differential amplifier is the difference in voltage between the positive and negative outputs. Therefore, the gain of a differential amplifier is as follows:

$$A_{vdiff} = \frac{v_{outp} - v_{outm}}{v_{inp} - v_{inm}}$$


Signal conventions

Notice that the actual voltages V_{outp} and V_{outm} are unimportant in this equation. Only the *difference* between these voltages counts. However, the actual values *must* stay between V_{DD} and V_{SS} .

The common mode of the output is $(V_{outp} + V_{outm})/2$. This is the average of the two output voltages.

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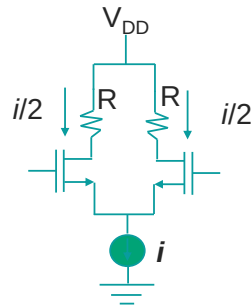
Setting Common-Mode Voltage

The output common mode of an amplifier can be determined by an open-loop circuit or by a closed-loop feedback mechanism.

A simple differential-pair amplifier with resistive load is an example of an open-loop mechanism to set output common mode:

Output common mode

$$V = V_{DD} - R \left(\frac{i}{2} \right)$$

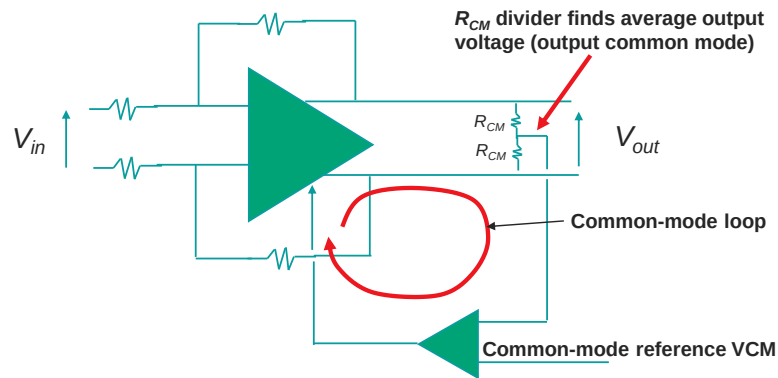


Resistive load diff pair

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Common-Mode Feedback

In more complicated differential amplifiers, a common-mode feedback circuit is required. The common-mode feedback loop must be analyzed for stability and frequency response in a similar manner to any other amplifier:



Note: Common-mode feedback loop maintains output common mode at V_{CM} .

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Discussion: Simple Common-Mode Example

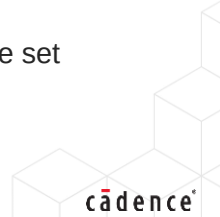
Advantages

- Simple circuit
- Very stable common-mode loop
- Low Power

Disadvantages

- Output common mode is not well defined over process corners.
- Some nonlinearity is introduced.
- Circuit does not work over the full range from VSS to VDD.

With low-supply voltages in modern processes, the common-mode voltage often must be set precisely to maximize the dynamic range.



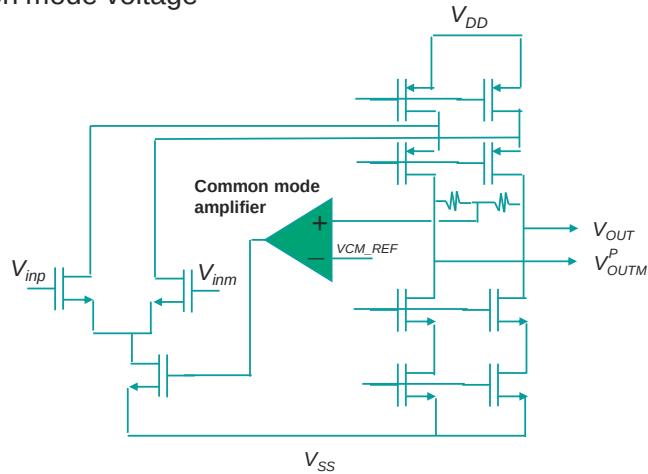
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Folded Cascode Amplifier with Common-Mode Loop

Advantage: precise control of common mode voltage

Disadvantages:

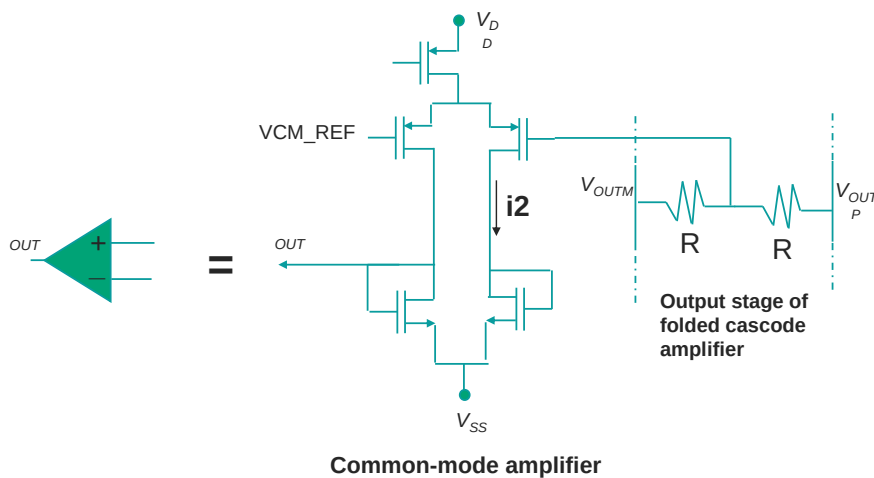
- Extra power required
- Area of amplifier
- Resistors limit open-loop gain
- Possible instability



Folded cascode amplifier with common-mode loop

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Example: Common-Mode Amplifiers



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Switched-Capacitor Common-Mode Feedback

The continuous-time common-mode feedback circuits in the previous slides can be replaced with a switched-capacitor common-mode feedback circuit.

Advantages

- Low power
- Does not reduce loop gain in switched-capacitor-amplifier applications

Disadvantages

- Requires a clock
- Usually not used in continuous-time applications due to noise injected from the clock

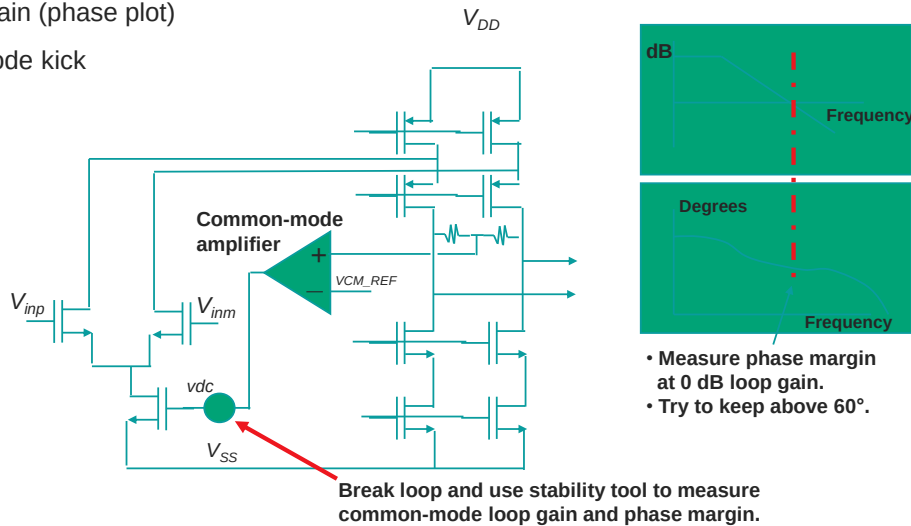


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Common-Mode Loop-Stability Measurements

Two methods are used:

- Traditional gain (phase plot)
- Common-mode kick

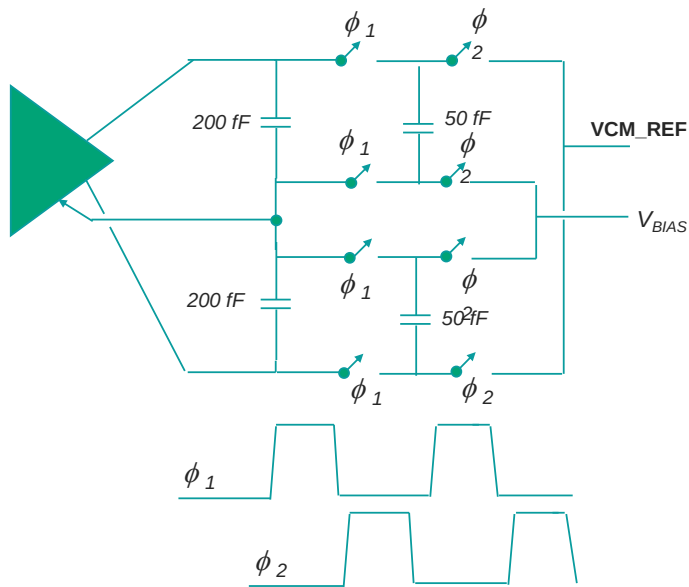


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Example: Switched-Capacitor Feedback Circuit



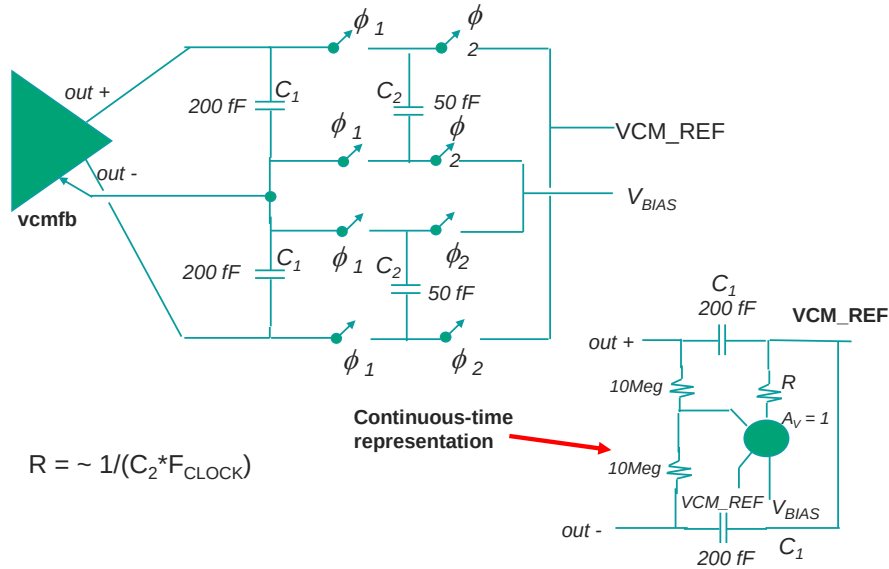
VCM_REF equals the desired common mode voltage at the output.

V_{BIAS} is the nominal voltage the amplifier requires to set the output to VCM.

ϕ_1 and ϕ_2 are nonoverlapping clocks.

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Continuous-Time Approximation



A continuous-time approximation of a switched-capacitor common-mode circuit

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Switched-Capacitor Common-Mode Discussion

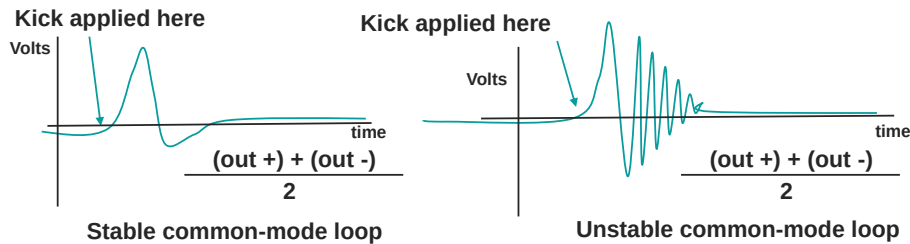
- Lower power than continuous time feedback circuit.
- Does not reduce loop gain of switched capacitor circuit.
- Requires a clock.
- Continuous time approximation circuit is used for AC simulations and to speed up transient simulations.
- Often a continuous-time-approximation circuit is used in initial simulations and then replaced with a switched-capacitor circuit for final transient simulations.



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Switched-Cap Common-Mode Circuit: Simulating Stability

- You cannot easily run an AC common-mode stability simulation with a switched-capacitor common-mode loop.
- One way to check the stability of a common-mode loop is to kick the common-mode loop in a transient simulation and observe the recovery characteristics.
- Often a current pulse is applied in the common-mode loop to supply the kick.
- Advanced simulators, such as SpectreRF, have so-called periodic AC analysis.



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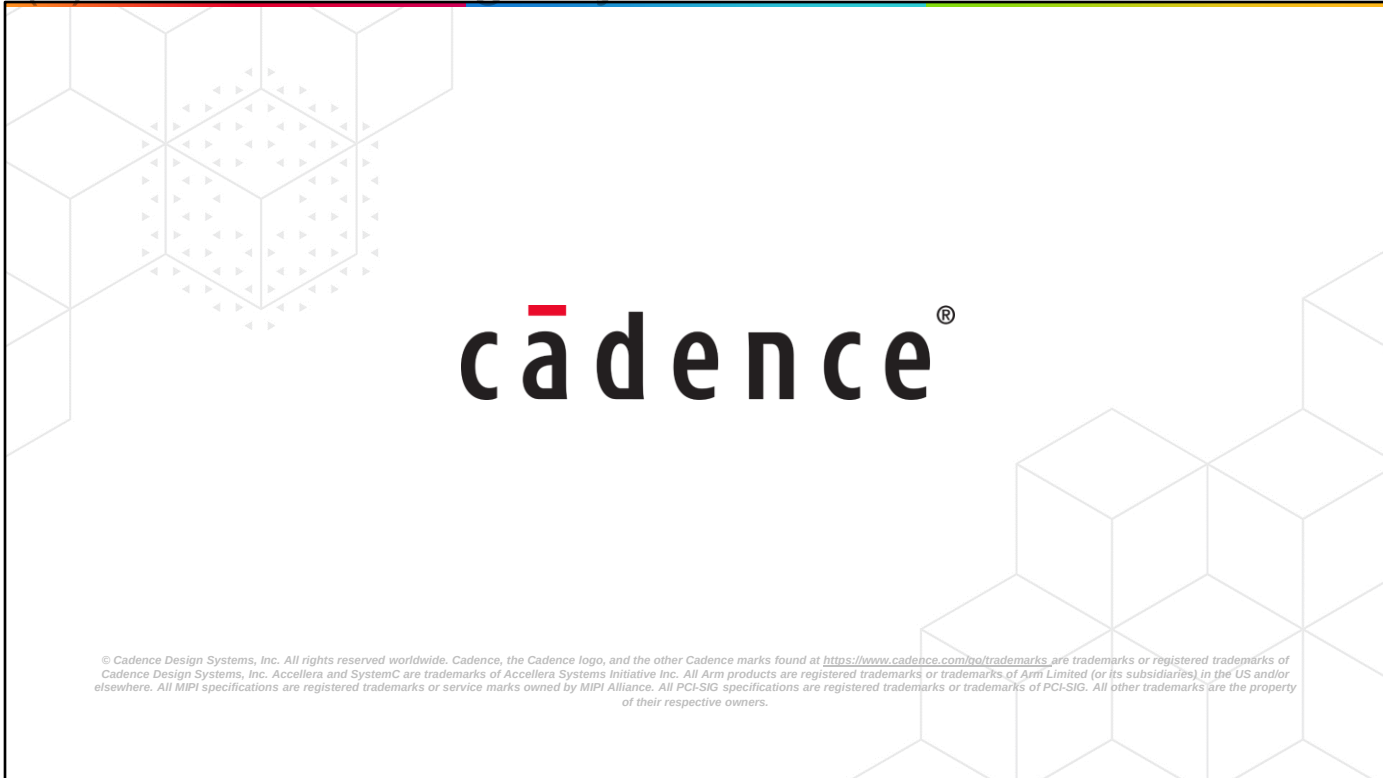
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Summary: Common-Mode Response

- All differential amplifiers have a common-mode response.
- You can set the common mode voltage several ways, which include
 - Common-mode feedback
 - Switched-capacitor feedback
- The switched-capacitor-feedback method requires that a simulation be done in the time domain to verify stability by giving the circuit a transient kick.



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